

AI Schema Builder Presentation

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AI Schema Builder

Intelligent Database Schema Generation for Backend Engineers

Powered by Google Gemini API (Free Tier)

Version 1.0 | 2025

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The Problem

■ Time-Consuming

Manually designing schemas for every table is slow and error-prone

■ Missed Relations

Foreign key relationships are often overlooked during early design

■ Poor Documentation

Schema docs are rarely kept in sync with actual database structure

■ Rework Cycles

Schema changes mid-project cause expensive refactoring cascades

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Our Solution

AI Schema Builder combines your domain knowledge with Gemini AI to generate production-ready database schemas in seconds – not hours.

1

Define Project

& Tables

2

Claude Generates

Schemas

3

Review

& Edit

4

Auto-Detect

FK References

5

Export

SQL / JSON

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Key Features

■ AI-Powered Generation

Claude generates columns, types, constraints, and indexes based on your table description

■ Interactive ERD

Live entity-relationship diagram updates as you add tables and accept FK relationships

■ Smart Reference Detection

Automatically suggests foreign key links across all tables in your project

■ Multi-Format Export

Export as SQL DDL, JSON schema, or Markdown documentation for your team

■ Secure API Key Mgmt

Encrypted storage of your Anthropic API keys – never shared, never logged

■ Schema Versioning

Every regeneration creates a snapshot – roll back or compare versions anytime

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Technical Architecture

Frontend

React + TypeScript + TailwindCSS + React Flow

Backend API

Node.js + Express + REST API

AI Layer (Free)

Google Gemini API (gemini-2.5-flash)

Database

PostgreSQL – Projects, Tables, Schemas, Versions

Infrastructure

Docker + Kubernetes | AWS ECS / GCP Cloud Run

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Google Gemini AI Integration

System Prompt Context

- Project Name & RDBMS type
- All existing table names
- Table descriptions

- Desired column types
- Relationship hints
- Response format: JSON

→

Gemini Returns

```
{ "columns": [  
  { "name": "id",  
    "type": "UUID",  
    "primaryKey": true },  
  { "name": "email",  
    "type": "VARCHAR(255)" }  
], "indexes": [...] }
```

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Deployment Strategy

Development

Docker Compose – local dev with hot-reload, local Postgres

Staging

AWS ECS / GCP Cloud Run – mirrors production, CI/CD validated

Production

Kubernetes (EKS/GKE) – autoscaling, managed DB, CDN-fronted

CI/CD Pipeline: GitHub Actions → Docker Build → Registry Push → ECS/K8s Deploy → Health Check

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Product Roadmap

Phase 1 – MVP

Weeks 1–4

✓ Auth + Projects

- ✓ Table CRUD

✓ Claude Schema Gen

- ✓ SQL Export

Phase 2 — Relations

Weeks 5–8

- ✓ FK Detection
- ✓ ERD Visualization

✓ Schema Versioning

- ✓ JSON Export

Phase 3 — Teams

Weeks 9–12

- ✓ Multi-user Projects
- ✓ Role-based Access
- ✓ Team API Keys

✓ Activity Logs

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Let's Build Smarter.

AI Schema Builder – reducing schema design time by 80%

github.com/your-org/ai-schema-builder

Questions?